

Module 52.1: Fixed-Income Bond Valuation: Prices and Yields

SCHWESERNOTES - BOOK 3

LOS 52.a: Calculate a bond's price given a yield-to-maturity on or between coupon dates.

Calculating the Value of an Annual Coupon Bond

The value of a coupon bond can be calculated by summing the present values of all of the bond's promised cash flows. The market discount rate appropriate for discounting a bond's cash flows is called the bond's **yield to maturity (YTM)**. If we know a bond's YTM, we can calculate its value, and if we know its value (market price), we can calculate its YTM.

Consider a newly issued 5-year, 10% coupon, annual-pay bond. For \$100 of par value, the coupon payments will be \$10 at the end of each year, and the \$100 par value will be paid at the end of Year 5. First, let's value this bond assuming that the appropriate discount rate (also called "yield") is 10%. The present value of the bond's cash flows discounted at 10% is as follows:

$$\frac{10}{(1.10)^1} + \frac{10}{(1.10)^2} + \frac{10}{(1.10)^3} + \frac{10}{(1.10)^4} + \frac{110}{(1.10)^5}$$

The calculator solution is as follows:

$$N = 5; PMT = 10; FV = 100; I/Y = 10; CPT \rightarrow PV = -100$$

where:

N = number of years

PMT = the *annual* coupon payment

I/Y = the *annual* discount rate (YTM)

FV = the par value or face value of the bond received at maturity

This calculation shows that when the coupon of a bond is equal to its yield, the bond's price is equal to par.

Professor's Note

Take note of a couple points here. The discount rate is entered as a whole number in a percentage, 10, not 0.10. The five coupon payments of \$10 each are taken care of in the N = 5 and PMT = 10 entries. The principal repayment is in the FV = 100 entry. Lastly, note that the PV is negative; it will be the opposite sign to the sign of PMT and FV. The calculator is just "thinking" that to receive the payments and future value (to own the bond), you must pay the present value of the bond today (you must buy the bond). That's why the PV amount is negative; it is a cash outflow to a bond buyer.

Now, let's value that same bond with a discount rate of 8%:

$$\frac{10}{(1.08)^1} + \frac{10}{(1.08)^2} + \frac{10}{(1.08)^3} + \frac{10}{(1.08)^4} + \frac{110}{(1.08)^5}$$

The calculator solution is as follows:

N = 5; PMT = 10; FV = 100; I/Y = 8; CPT → PV = -107.99

If the market discount rate for this bond were 8%, it would sell at a **premium** of \$7.99 above its par value. When bond yields decrease, the present value of a bond's payments, its market value, increases. Here, we also see that a bond with a coupon greater than its yield will be trading above par.

If we discount the bond's cash flows at 12%, this is the present value of the bond:

$$\frac{10}{(1.12)^1} + \frac{10}{(1.12)^2} + \frac{10}{(1.12)^3} + \frac{10}{(1.12)^4} + \frac{110}{(1.12)^5}$$

The calculator solution is as follows:

N = 5; PMT = 10; FV = 100; I/Y = 12; CPT → PV = -92.79

If the market discount rate for this bond were 12%, it would sell at a **discount** of \$100 - \$92.79 = \$7.21 to its par value. When bond yields increase, the present value of a bond's payments, its market value, decreases. Here, we also see that a bond with a coupon less than its yield will be trading below par.

Professor's Note

It's worth noting here that a 2% decrease in YTM increases the bond's value by more (\$7.99) than a 2% increase in yield decreases the bond's value (\$7.21). This illustrates that the bond's price-yield relationship is convex, as we will explain in more detail in a later reading.

Calculating the Value of a Semiannual Coupon Bond

Let's calculate the value of the same bond with semiannual payments. Rather than \$10 per year, the security will pay \$5 every six months. Assuming an annual YTM of 8%, we need to discount the coupon payments at 4% per period, which results in this present value:

$$\frac{5}{(1.04)^1} + \frac{5}{(1.04)^2} + \frac{5}{(1.04)^3} + \dots + \frac{5}{(1.04)^9} + \frac{105}{(1.04)^{10}}$$

The calculator solution is as follows:

N = 10; PMT = 5; FV = 100; I/Y = 4; CPT → PV = -108.11

The stated annualized YTM is equal to the periodic return of the bond multiplied by the number of periods in the year. In this case, the bond actually earns 4% every six months, hence the stated annualized YTM = 4% × 2 = 8%.

Professor's Note

One of our most frequently asked questions is, "For semiannual-pay bonds, why do we divide YTM by 2 instead of using $(1 + \text{YTM})^{1/2} - 1$?" We do this because a *YTM is not an effective annual rate*, unless it's an

annual-pay bond. YTM is an effective rate for the coupon period, times the number of coupon periods in a year.

- For a semiannual-pay bond, YTM is an effective semiannual rate times 2 periods.
- For a quarterly-pay bond, YTM is an effective quarterly rate times 4 periods.
- For a monthly-pay bond, YTM is an effective monthly rate times 12 periods.

We would expect the Level I exam to mostly feature annual-pay or semiannual-pay bonds.

Calculating Yield to Maturity

Now let's calculate the stated yield of the same bond after changes in market conditions have caused the bond price to move to 105. (This price is stated as a percentage of par quote—in other words, investors will pay 105% of the par value they purchase. The easiest way to interpret this price is as the price of \$100 of par.)

We solve for the semiannual return of the bond using the following calculator inputs:

$$N = 10; PMT = 5; FV = 100; PV = -105; CPT \rightarrow I/Y = 4.37\%$$

This is the true semiannual return of the bond. To quote a YTM, we need to multiply by two to give a stated YTM of $4.37\% \times 2 = 8.74\%$. Also note the negative sign for the PV input: we must respect the direction of cash flows (negative for the cash outflow today, positive for the cash inflow later), or else the calculator will not be able to calculate a rate of return.

To actually earn the YTM over the life of a bond, the investor must hold the bond to maturity, the issuer must make all the promised payments, and the investor must be able to reinvest the periodic cash flows and earn the same YTM.

Accrued Interest, Flat Price, and Full Price

So far we have been calculating bond values on the date a coupon is paid, as the present value of the remaining coupons. For most actual bond trades, the settlement date, which is when cash is exchanged for the bond, will fall between coupon payment dates.

Bond pricing has to account for the fact that the next coupon will be paid to the buyer, but a portion of it (the **accrued interest**) is owed to the seller. The accrued interest since the last payment date can be calculated as the coupon payment times the portion of the coupon period that has passed between the last coupon payment date and the settlement date of the transaction. That is:

$$\text{accrued interest} = \text{coupon payment} \times \frac{\text{days from last coupon to settlement}}{\text{days in coupon period}}$$

Financial markets use a variety of **day count methods**. Two of the most common are the **actual/actual convention**, which uses the actual number of days between coupon payments and the actual number of days between the last coupon date and the settlement date; and the **30/360 convention**, which assumes each month has 30 days and a year has 360 days. Government bonds typically use actual/actual, while corporate bonds typically use 30/360.

EXAMPLE: ACCRUED INTEREST

An investor buys a 4% annual-pay bond that pays its coupons on May 15. The investor's order settles on August 10. Calculate the accrued interest that is owed to the bond seller, using the 30/360 method and the actual/actual method.

Answer:

The annual coupon payment is $4\% \times \$100 = \4 .

Using the 30/360 method, interest is accrued for $30 - 15 = 15$ days in May; 30 days each in June and July; and 10 days in August, or $15 + 30 + 30 + 10 = 85$ days:

$$\text{accrued interest (30/360 method)} = \frac{85}{360} \times \$4 = \$0.944$$

Using the actual/actual method, interest is accrued for $31 - 15 = 16$ days in May; 30 days in June; 31 days in July; and 10 days in August, or $16 + 30 + 31 + 10 = 87$ days.

$$\text{accrued interest (actual/actual method)} = \frac{87}{365} \times \$4 = \$0.953$$

Bond prices are quoted without accrued interest. This is because, holding yield constant, including accrued interest would make a bond's price appear to increase on each day of a coupon period and drop suddenly on the coupon payment date. A bond's quoted price is known as its **flat price** (or clean price).

A bond's **full price** (also known as its invoice price or dirty price) is the sum of its flat price and the accrued interest. However, we cannot simply calculate a flat price and add accrued interest to it. Instead we must calculate the full price and derive the flat price from it:

$$\text{flat price} = \text{full price} - \text{accrued interest}$$

The method for calculating the full price is as follows:

Step 1: Calculate the value of the bond on the last coupon date.

Step 2: Compound this value at the YTM per period, over the number of days since the last coupon payment:

$$\text{full price} = \text{PV on last coupon date} \times \left(1 + \frac{\text{YTM}}{\text{periods per year}} \right)^{\frac{\text{days since last coupon}}{\text{days in coupon period}}}$$

Let's work an example for a specific bond.

EXAMPLE: CALCULATING THE FULL AND FLAT PRICES OF A BOND

A 5% bond makes coupon payments on June 15 and December 15, and is trading with a YTM of 4%. The bond is purchased, and will settle on August 21 when there will be four coupons remaining until maturity. Calculate the full price, accrued interest, and the flat price of the bond using actual days.

Answer:

Step 1: Calculate the value of the bond on the last coupon date (coupons are semiannual, so we use $4 / 2 = 2\%$ for the periodic discount rate):

$$N = 4; \text{PMT} = 2.5; \text{FV} = 100; \text{I/Y} = 2; \text{CPT} \rightarrow \text{PV} = -101.904$$

Step 2: Adjust for the number of days since the last coupon payment:

days between June 15 and December 15 = 183 days

days between June 15 and settlement on August 21 = 67 days

$$\text{full price} = 101.904 \times (1.02)^{67/183} = 102.645$$

Accrued interest on the settlement date of August 21 is as follows:

$$\$2.5 \times (67 / 183) = \$0.915$$

$$\text{Flat price} = 102.645 - 0.915 = 101.73.$$

Note that the flat price is *not* the present value of the bond on its last coupon payment date, $101.73 < 101.904$.

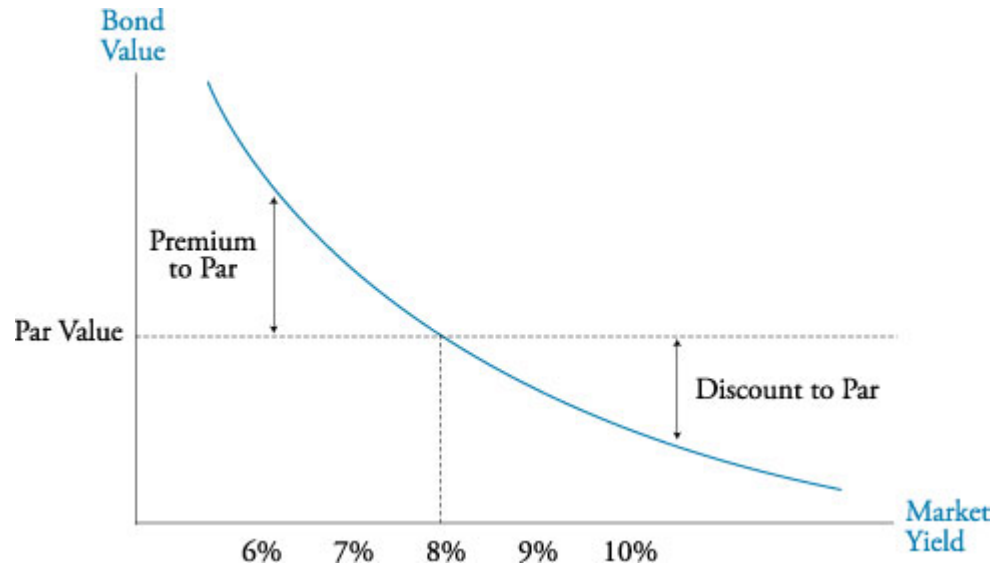
LOS 52.b: Identify the relationships among a bond's price, coupon rate, maturity, and yield-to-maturity.

We can summarize the relationships between price and specific bond features as follows:

1. At a point in time, a decrease (increase) in a bond's YTM will increase (decrease) its price. That is, there is an inverse relationship between yield and price.
2. Other things equal, the price of a bond with a lower coupon rate is more sensitive to a change in yield than is the price of a bond with a higher coupon rate.
3. Other things equal, the price of a bond with a longer maturity is more sensitive to a change in yield than is the price of a bond with a shorter maturity.
4. The percentage decrease in value when the YTM increases by a given amount is smaller than the increase in value when the YTM decreases by the same amount (the price-yield relationship is convex).

Market Yield vs. Bond Value for an 8% Coupon Bond illustrates the convex relationship between a bond's price and its YTM.

Market Yield vs. Bond Value for an 8% Coupon Bond



Relationship Between Price and Maturity

Before maturity, a bond can be selling at a significant discount or premium to par value. However, regardless of its required yield, the price will converge to par value as maturity approaches. Consider a bond with a 3-year life paying 6% semiannual coupons. The bond values corresponding to required yields of 3%, 6%, and 12% as the bond approaches maturity are presented in **Bond Values and the Passage of Time**.

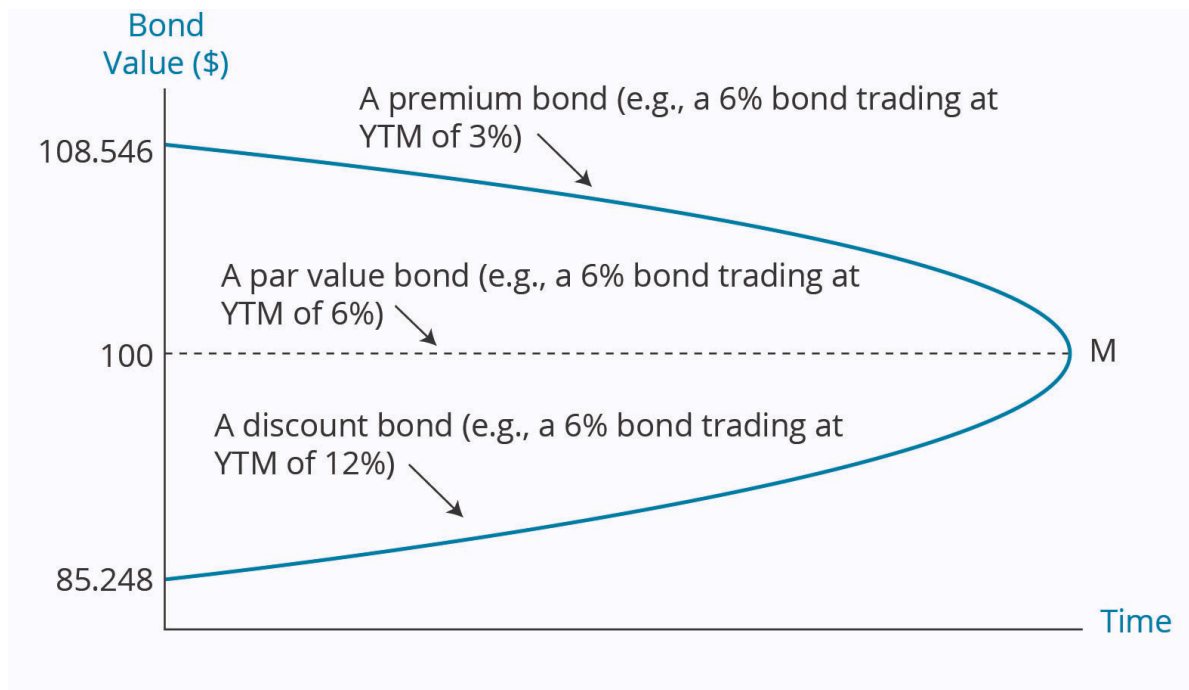
Bond Values and the Passage of Time

Time to Maturity(In Years)	YTM = 3%	YTM = 6%	YTM = 12%
3.0	108.546	100	85.248

2.5	107.174	100	87.363
2.0	105.782	100	89.605
1.5	104.368	100	91.981
1.0	102.934	100	94.500
0.5	101.478	100	97.169
0.0	100.000	100	100.000

The change in value associated with the passage of time for the three bonds represented in **Bond Values and the Passage of Time** is presented graphically in **Premium, Par, and Discount Bonds**. This convergence to par value ("pull to par") at maturity is known as the **constant-yield price trajectory** because it shows how the bond's price would change as time passes if its YTM remained constant.

Premium, Par, and Discount Bonds



LOS 52.c: Describe matrix pricing.

Matrix pricing is a method of estimating the required YTM (or price) of bonds that are currently not traded, or infrequently traded. The procedure is to use the YTM of traded bonds that have credit quality very close to that of a nontraded or infrequently traded bond and are similar in maturity and coupon, to estimate the required YTM.

EXAMPLE: PRICING AN ILLIQUID BOND

Rob Phelps, CFA, is estimating the value of a nontraded 4% annual-pay, A+ rated bond that has three years remaining until maturity. He has obtained the following yields to maturity on similar corporate bonds:

- A+ rated, 2-year annual-pay, YTM = 4.3%
- A+ rated, 5-year annual-pay, YTM = 5.1%
- A+ rated, 5-year annual-pay, YTM = 5.3%

Estimate the value of the nontraded bond.

Answer:

Step 1: Take the average YTM of the 5-year bonds:

$$(5.1 + 5.3) / 2 = 5.2\%$$

Step 2: Interpolate the 3-year YTM based on the 2-year and average 5-year YTM:

$$4.3\% + (5.2\% - 4.3\%) \times [(3 \text{ years} - 2 \text{ years}) / (5 \text{ years} - 2 \text{ years})] = 4.6\%$$

Step 3: Price the nontraded bond with a YTM of 4.6%:

$$N = 3; PMT = 4; FV = 100; I/Y = 4.6; CPT \rightarrow PV = -98.354$$

The estimated value is \$98.354 per \$100 par value.

In Step 2 in the preceding example, we have used simple linear interpolation. Because the maturity of the nontraded bond is three years, we estimate the YTM on the 3-year bond as the yield on the 2-year bond, plus one-third of the difference between the YTM of the 2-year bond and the average YTM of the 5-year bonds. (The difference between the maturities of the 2-year bond and the 3-year bond is one year, and the difference between the maturities of the 2-year and 5-year bonds is three years.)

A variation of matrix pricing used for new bond issues focuses on spreads. The required yield spread to a benchmark for a new issue bond can be estimated by observing spreads on existing similar securities, as demonstrated in the following example.

EXAMPLE: ESTIMATING THE SPREAD FOR A NEW 6-YEAR, A RATED BOND ISSUE

Consider the following market yields:

- 4-year, U.S. Treasury bond, YTM 1.48%
- 5-year, A rated corporate bond, YTM 2.64%
- 6-year, U.S. Treasury bond, YTM 2.15%

Estimate the required yield spread on a newly issued 6-year, A rated corporate bond.

Answer:

We will use the existing 5-year, A rated corporate bond to estimate the required yield spread of the issuer by comparing the YTM of the 5-year corporate bond to the interpolated 5-year Treasury bond YTM.

Interpolated 5-year Treasury bond YTM:

$$\begin{aligned} &= 1.48\% + (2.15\% - 1.48\%) \times [(5 \text{ years} - 4 \text{ years}) / (6 \text{ years} - 4 \text{ years})] \\ &= 1.815\% \end{aligned}$$

Note: Because the target maturity of the existing bond (5 years) is midway between the two Treasury bond maturities (4 and 6 years), we could have simply averaged the two Treasury bond yields here $(1.48\% + 2.15\%) / 2 = 1.815\%$.

The yield spread on existing 5-year corporate debt is $2.64\% - 1.815\% = 0.825\%$.

We will apply this yield spread to the new 6-year corporate debt issue:

YTM for the new 6-year corporate bond = $2.15\% + 0.825\% = 2.975\%$

Reading 52: Key Concepts

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LOS 52.a

The price of a bond is the present value of its future cash flows, discounted at the bond's yield to maturity.

For an annual coupon bond with N years to maturity:

$$\text{price} = \frac{\text{coupon}}{(1+\text{YTM})} + \frac{\text{coupon}}{(1+\text{YTM})^2} + \dots + \frac{\text{coupon} + \text{principal}}{(1+\text{YTM})^N}$$

For a semiannual coupon bond with N years to maturity:

$$\text{price} = \frac{\text{coupon}}{\left(1 + \frac{\text{YTM}}{2}\right)} + \frac{\text{coupon}}{\left(1 + \frac{\text{YTM}}{2}\right)^2} + \dots + \frac{\text{coupon} + \text{principal}}{\left(1 + \frac{\text{YTM}}{2}\right)^{N \times 2}}$$

The full price of a bond includes interest accrued between coupon dates. The flat price of a bond is the full price minus accrued interest.

Accrued interest for a bond transaction is calculated as the coupon payment times the portion of the coupon period from the previous payment date to the settlement date.

Methods for determining the period of accrued interest include actual days or 30-day months and 360-day years.

LOS 52.b

A bond's price and YTM are inversely related. An increase in YTM decreases the price, and a decrease in YTM increases the price.

Bond prices are convex with respect to yield movements, which means price increases when yields fall are greater in magnitude than the fall in prices caused by an equivalent yield rise.

A bond will be priced at a discount to par value if its coupon rate is less than its YTM (deficient coupon), and at a premium to par value if its coupon rate is greater than its YTM (excessive coupon).

Prices are more sensitive to changes in YTM for bonds with lower coupon rates and longer maturities, and less sensitive to changes in YTM for bonds with higher coupon rates and shorter maturities.

A bond's price moves toward par value as time passes and maturity approaches.

LOS 52.c

Matrix pricing is a method used to estimate the yield to maturity for bonds that are not traded or infrequently traded. The yield is estimated based on the yields of traded bonds with the same credit quality. If these traded bonds have different maturities than the bond being valued, linear interpolation is used to estimate the subject bond's yield.